

E-Commerce Sales Analysis Internship Project Report

Project Overview

This project analyzes e-commerce sales data using Python to identify sales trends, top-performing products, regional performance, and business insights through data visualization.

Introduction

The **E-Commerce Sales Analysis** project aims to analyze sales data from an online retail business using Python. The project focuses on cleaning raw sales data, performing exploratory data analysis (EDA), and generating visualizations to identify sales trends, product performance, and regional revenue distribution.

The objective is to transform raw business data into meaningful insights that can support decision-making and improve overall business performance.

Goals

- Understand sales trends over time.
- Identify top-performing products.
- Compare sales across different regions.
- Analyze quantity sold by product.
- Generate actionable business insights.
- Create professional visualizations for reporting.

Objectives

- Analyze sales trends
- Identify top-selling products
- Compare regional sales
- Generate business insights
- Create visualizations

- **Dataset Description**

Column	Description
Date	Date of Sale
Product	Product Name
Region	Sales Region
Quantity	Units Sold
Price	Unit Price
Sales	Total Sales Amount

Setup Instructions

1. Install Python 3.x
2. Install pandas, numpy, matplotlib
3. Open Analysis.ipynb in Jupyter Notebook
4. Run all cells

Project Execution Steps

1. Download the project folder.
2. Open the folder in VS Code or Jupyter Notebook.
3. Install all required libraries.
4. Open **Analysis.ipynb**.
5. Run all notebook cells.
6. Charts will be generated automatically.

Code Structure

data/
visualizations/
Analysis.ipynb
README.md
requirements.txt
report/

Technical Details

Libraries: Pandas, NumPy, Matplotlib.

Workflow: Load Data → Clean Data → Analyze → Visualize → Generate Insights.

Visual Documentation

Project Workflow

CSV Dataset



Load Data



Data Cleaning



Exploratory Data Analysis



Visualization

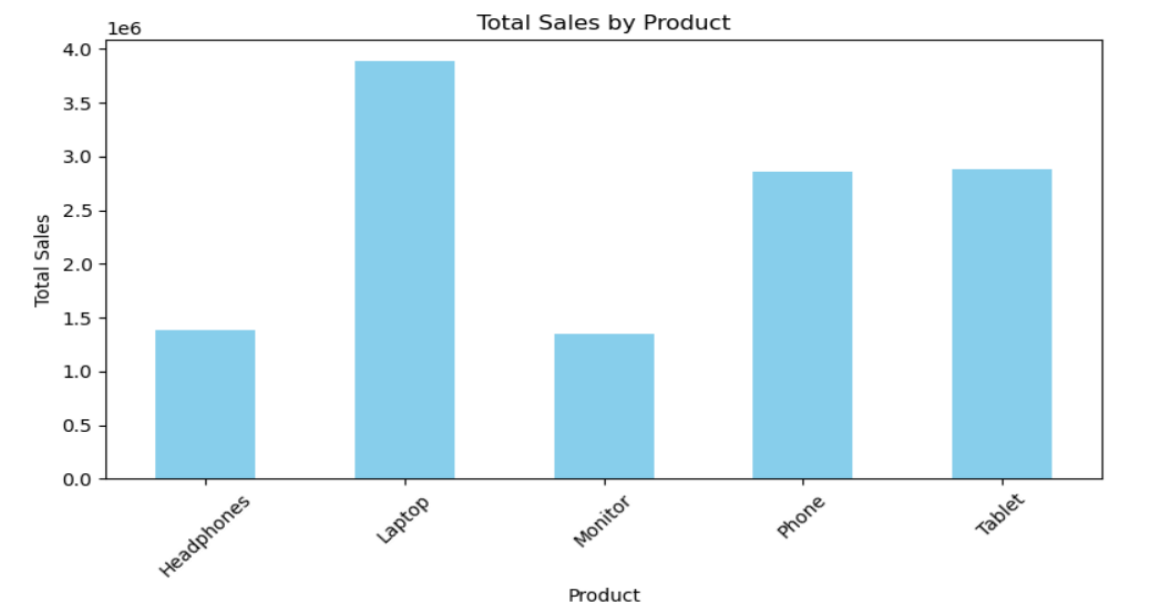


Business Insights

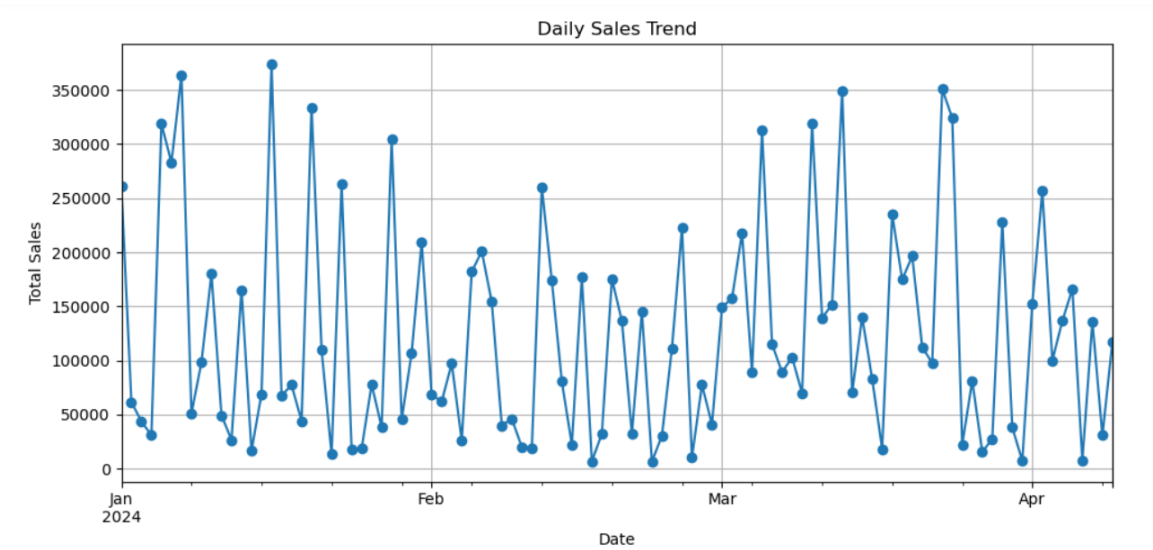


Final Report

Total Sales by Product

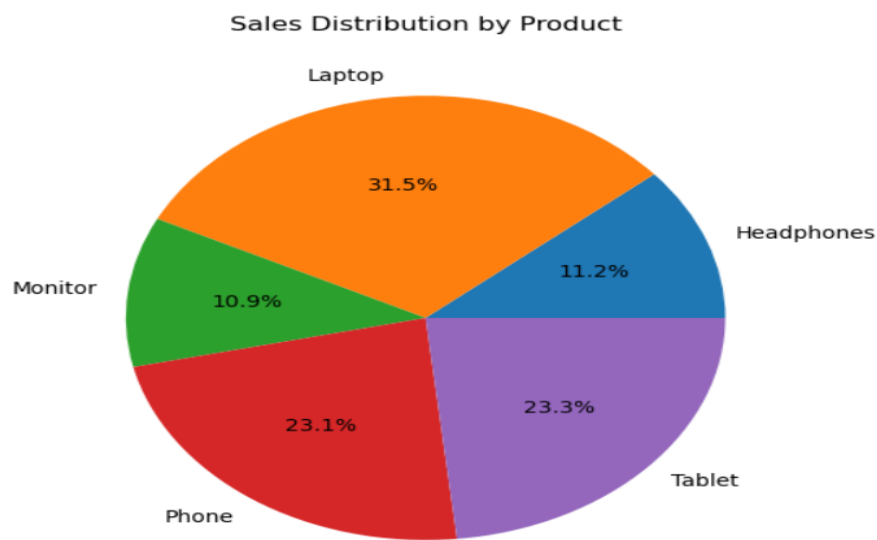


Daily Sales Trend

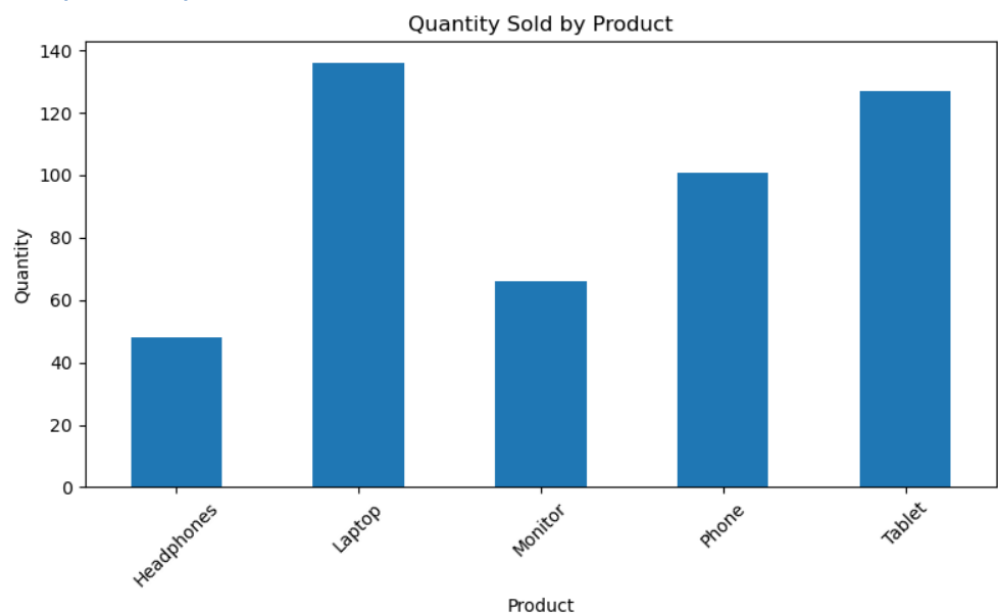


Observation: Daily Sales Trend chart generated successfully.

Sales Distribution by Product

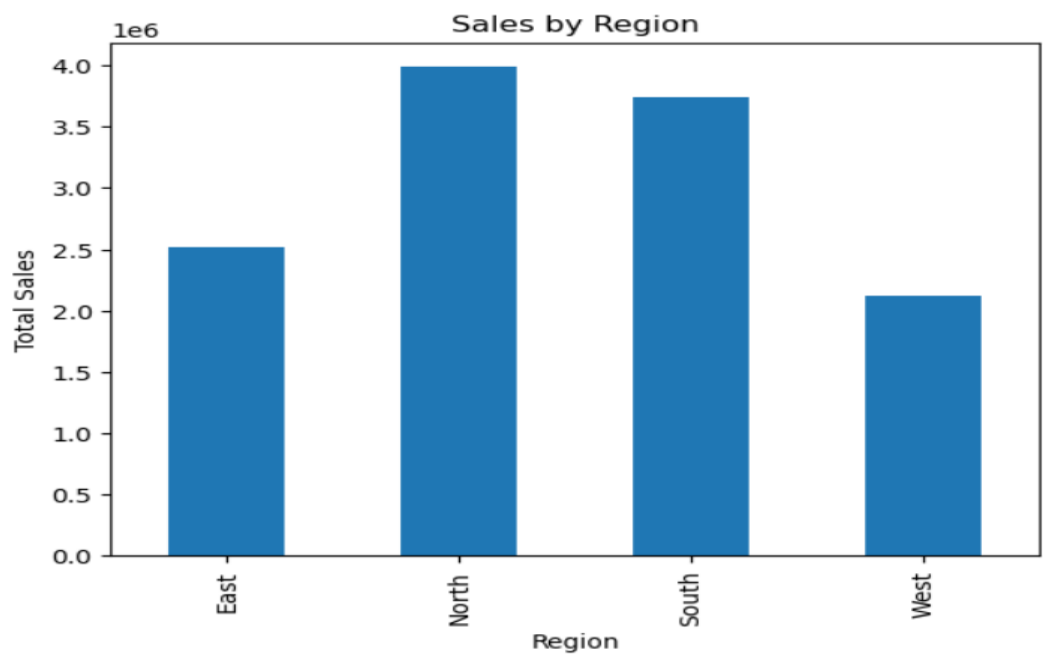


Quantity Sold by Product



Observation: Quantity Sold by Product chart generated successfully.

Sales by Region



Observation: Sales by Region chart generated successfully.

Business Insights

The analysis revealed several valuable insights:

- Laptop is the highest revenue-generating product.
- North region has the strongest sales performance.
- Daily sales fluctuate due to varying customer demand.
- Tablet also maintains consistently high sales.
- Increasing inventory for top-performing products may improve profitability.
- Marketing campaigns can be focused on high-performing regions.

Testing Evidence

Test Case	Result	Status
Dataset Loading	Success	Pass
Missing Values	Validated	Pass
Charts	Generated	Pass

Conclusion

This project successfully demonstrates an end-to-end data analytics workflow using Python. The dataset was cleaned, analyzed, and visualized to identify product performance, regional sales, and customer purchasing trends. The generated visualizations and business insights support informed decision-making and showcase practical data analysis skills suitable for internship evaluation.

Future Scope

- Develop an interactive dashboard using Power BI or Tableau.
- Integrate real-time sales data.
- Apply machine learning for sales forecasting.
- Perform customer segmentation.
- Automate report generation.

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